The natural history of untreated pulmonary tuberculosis in adults: a systematic review and meta-analysis

Bianca Sossen, Alexandra S Richards*, Torben Heinsohn, Beatrice Frascella, Federica Balzarini, Aurea Oradini-Alacreu, Anna Odone, Ewelina Rogozinska, Brit Häcker, Frank Cobelens, Katharina Kranzer, Rein M G J Houben†, Hanif Esmail†*

Stages of tuberculosis disease can be delineated by radiology, microbiology, and symptoms, but transitions between these stages remain unclear. In a systematic review and meta-analysis of studies of individuals with untreated tuberculosis who underwent follow-up (34 cohorts from 24 studies, with a combined sample of 139 063), we aimed to quantify progression and regression across the tuberculosis disease spectrum by extracting summary estimates to align with disease transitions in a conceptual framework of the natural history of tuberculosis. Progression from microbiologically negative to positive disease (based on smear or culture tests) in participants with baseline radiographic evidence of tuberculosis occurred at an annualised rate of 10% (95% CI 6·2–13·3) in those with chest x-rays suggestive of active tuberculosis, and at a rate of 1% (0·3–1·8) in those with chest x-ray changes suggestive of inactive tuberculosis. Reversion from microbiologically positive to undetectable disease in prospective cohorts occurred at an annualised rate of 12% (6·8–18·0). A better understanding of the natural history of pulmonary tuberculosis, including the risk of progression in relation to radiological findings, could improve estimates of the global disease burden and inform the development of clinical guidelines and policies for treatment and prevention.

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*Contributed equally

†Contributed equally

Department of Medicine, Faculty of Health Sciences (B Sossen MBChB), **and Wellcome Centre for Infectious Diseases Research in Africa, Institute of Infectious Diseases and Molecular Medicine** (H Esmail PhD), **University of Cape Town, Cape Town, South Africa; Institute for Global Health** (B Sossen, T Heinsohn BMBCh, H Esmail) **and MRC Clinical Trials Unit** (E Rogozinska PhD, H Esmail), **UCL, London, UK; TB Modelling Group, TB Centre** (A S Richards MMathPhys, Prof R M G J Houben PhD), **Department of Infectious Disease Epidemiology, Faculty of Epidemiology and Public Health** (A S Richards, Prof R M G J Houben), **and Clinical Research Department, Faculty of Infectious and Tropical Disease** (Prof K Kranzer PhD), **London School of Hygiene and Tropical Medicine, London, UK; Department of Epidemiology, Helmholtz Centre for Infection Research, Braunschweig, Germany** (T Heinsohn); **School of Public Health, Vita-Salute San Raffaele University, Milan, Italy** (B Frascella MD, F Balzarini MD, A Oradini-Alacreu MD); **Department of Public Health, Experimental and Forensic Medicine, University of Pavia, Pavia, Italy** (Prof A Odone PhD); **German Central Committee against Tuberculosis (DZK), Berlin, Germany** (B Häcker MD); **Department of Global Health,**

Introduction

Although there has been clinical awareness of tuberculosis for centuries, its natural history remains incompletely understood. Characterisation of tuberculosis has varied between separating it into binary states of latent infection and active disease, and considering it as a condition that exists on a dynamic continuum.[1–4] In the early 20th century, tuberculosis control relied on the early identification of people with evidence of disease, particularly with chest x-ray screening. Researchers were able to highlight the heterogeneity and dynamics of disease evolution between individuals through longitudinal assessment.[5–8] With the discovery of effective treatment in the mid-20th century and driven by the need for scalable, programmatic treatment algorithms, a binary description of disease states reflecting two extremes (latent infection and active disease) became established.[9] Although this provided a useful paradigm, the more nuanced understanding of disease natural history was arguably forgotten.

An accurate understanding of the kinetics of tuberculosis natural history is now increasingly crucial at both individual and population levels, with implications for disease management, population-level prevention and control, and estimation of disease burden. Treatment of patients whose disease states fall between latent and active tuberculosis—eg, those with abnormalities suggestive of active disease on x-ray but who are negative on microbiological testing—is not adequately covered by management algorithms, but progress could be driven by improved understanding of the risk of disease progression. A better understanding of the natural history is also a priority for vaccine development.[10] Additionally, estimates of tuberculosis incidence currently rely strongly on assumptions about the progression, regression, and mortality of untreated tuberculosis, of which only mortality estimates are informed by systematic reviews of available literature.[11–13] Furthermore, estimation methods do not account for patients having different stages of tuberculosis, which can be detected in disease prevalence surveys, including individuals who have culture-positive disease but a negative symptom screen (referred to as subclinical) or those with x-rays suggestive of tuberculosis.[14] Given the implications for health-care seeking and the potential for interrupting or preventing transmission, a better understanding of this natural history is key to informing estimates of tuberculosis disease burden and policies for care and prevention.

Across the disease continuum, key stages in the evolution of pulmonary tuberculosis can be marked by diagnostic tests that have been available for over a century, allowing categorisation within a widely accepted conceptual framework (figure 1).[1,2,14–16] The emergence of disease pathology is generally first visible as typical radiographic features, with differing sensitivity according to the radiographic tool used. Microbiological detection in sputum signals the presence of bacilli (and potential infectiousness), and the reporting of symptoms marks the development of active, clinical disease. Transitions across all these stages can be fully studied only in the absence of treatment and hence can no longer be ethically investigated. We did a systematic review and meta-analysis focusing on studies from the pre-chemotherapy era to determine which of these transitions could be adequately described by the existing literature and to provide parameters for the rate of progression and regression along the disease spectrum.

Methods

Search strategy and selection criteria

The protocol for this systematic review and meta-analysis is registered with PROSPERO (CRD42019152585). The study is reported in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) guidelines.[17] We searched for articles

Summary of systematic review and meta-analysis

Aim

We sought to quantify progression and regression across the spectrum of tuberculosis disease by systematically reviewing studies of individuals who were screened for or confirmed to have tuberculosis and followed up for at least 12 months without treatment.

Methods

We searched PubMed, EMBASE, and Web of Science from database inception to Dec 31, 1960; we searched Index Medicus from Jan 1, 1903, to Dec 31, 1945, and reviewed the extensive collections of the authors without date restrictions. Reports published in English and German were considered. Eligible studies were observational cohorts and clinical trials that included adolescents or adults with tuberculosis or recent tuberculosis exposure who were followed up for at least 12 months without therapeutic intervention. Two authors independently reviewed titles, abstracts, and full-text reports for inclusion. Quality was assessed with a modified Newcastle–Ottawa Scale score; highly biased studies were excluded. Summary estimates were extracted to align with tuberculosis disease transitions in a conceptual model, and we used meta-analysis of proportions with random-effects to synthesise the extracted data. The study is registered with PROSPERO (CRD42019152585).

Findings

Titles and abstracts for 10 623 studies were screened and full texts for 1503 studies reviewed. 223 studies met inclusion criteria. 109 studies were excluded because of a high risk of bias and 90 did not have extractable data. 24 studies (34 cohorts with a combined sample of 139 063) were included. Progression from microbiologically negative to positive disease in those with radiographic tuberculosis evidence occurred at an annualised rate of 10% (95% CI 6·2–13·3) in those with chest x-rays suggestive of active tuberculosis and at a rate of 1% (0·3–1·8) in those with chest x-rays suggestive of inactive tuberculosis. Reversion from microbiologically positive to undetectable disease in prospective cohorts occurred at an annualised rate of 12% (6·8–18·0). Studies reported symptoms poorly, precluding direct estimation of transitions from subclinical (asymptomatic, culture-positive) disease.

Implications

Our estimates of the rate of progression in people with radiographic evidence of disease and the rate of self-cure in those with microbiologically positive disease could inform estimates of the global burden of tuberculosis disease, clinical guidelines for the prevention and management of disease, and policy decisions on the need for interventions in people identified to have radiographic evidence of disease at screening.

from the pre-chemotherapy era using a combination of electronic and manual searches. We did electronic searches of MEDLINE (via PubMed), EMBASE, and Web of Science for articles published from database inception (1946, 1947, and 1900, respectively) to Dec 31, 1960, in two languages with high yield for study designs of interest in this period: English and German. Additionally, we manually searched the titles of papers in Index Medicus published between Jan 1, 1903 and Dec 31, 1945; volumes from 1895–1902 were not available. The systematic search was restricted to manuscripts published up to 1960 to include cohorts observed in the pre-chemotherapy era while allowing for a publication delay of earlier cohorts. Furthermore, relevant articles from the authors' own collections, and references from those articles and from key review articles, that met the criteria for data extraction were included. Authors' collections and references identified from selected articles were searched without date restrictions.

Amsterdam University Medical Centers, University of Amsterdam, Amsterdam, Netherlands (Prof F Cobelens PhD); **Amsterdam Institute for Global Health and Development, Amsterdam, Netherlands** (Prof F Cobelens); **Biomedical Research and Training Institute, Harare, Zimbabwe** (Prof K Kranzer); **Division of Infectious Diseases and Tropical Medicine, University Hospital, LMU Munich, Munich, Germany** (Prof K Kranzer)

Correspondence to: Dr Hanif Esmail, MRC Clinical Trials Unit at UCL, 90 High Holborn, London WC1V 6LJ, UK **h.esmail@ucl.ac.uk**

See **Online** for appendix

For the electronic search terms, we used both modern and historical terminology in English and German (see appendix pp 3–5 for full search strategies). All titles were imported into Covidence systematic review software (Veritas Health Innovation, Melbourne, Australia). After removal of duplicates, all titles and abstracts were screened for relevance by two independent reviewers, with a third reviewer resolving conflicts (BS, ASR, BF, FB, AO-A, TH, RMGJH, or HE in English; TH, BH, or KK in German). We searched for full-text articles in the online library stores of the Wellcome library and the British Library (English articles), and the library of the German Central Committee against Tuberculosis and the German Tuberculosis Archive (German articles), and in online archive websites (eg, the HathiTrust Digital Library and the US Internet Archive). If manuscripts could not be found through any of these sources, they were not included. Two independent reviewers reviewed all full-text articles for eligibility (BS, ASR, BF, FB, AO-A, or TH in English; TH, BH, or KK in German). Articles were included if they presented a longitudinal cohort of at least 25 adolescents (≥10 years old), adults, or both, that was followed up (radiologically, microbiologically, and clinically) for at least 12 months from identification of one of the following: (1) a positive tuberculin skin test following recent tuberculosis exposure; (2) radiographic abnormalities suggestive of tuberculosis; or (3) positive microbiology for tuberculosis (smear microscopy, mycobacterial culture, or both). A minimum of 12 months was selected to ensure an adequate number of events. Articles were excluded if they made no attempt to confirm tuberculosis disease microbiologically, if they presented no new data (eg, review articles), if all participants received a therapeutic (medical or surgical) intervention or the data from those who did not receive a therapeutic intervention could not be extracted, or if 5% or more of the cohort was under 10 years old and data for these children could not be separated from adolescent or adult data.

All eligible articles were assessed for risk of bias with an adapted Newcastle–Ottawa Scale (NOS), which uses a star scoring system, with a maximum of seven stars (general quality assessment), by two reviewers per language (appendix p 6); conflicts were resolved by consensus (BS, ASR, BF, FB, AO-A, TH, RMGJH, or HE in English; TH, BH, or KK in German). To pass the general quality assessment, studies could lose only two stars in the study selection and outcome domains of the NOS. The comparability domain was not assessed as this systematic review did not use control groups. An additional quality assessment tool was designed for this systematic review to assess the quality of specific diagnostic compartments in the study cohorts—ie, radiological, microbiological, and symptom status (appendix pp 7–8). Although this specific quality assessment gave a sense of quality of the study designs, we did not use it to inform study eligibility. Studies that passed the NOS assessment were extracted by one reviewer using a standardised electronic tool designed in Excel, and data points were confirmed by a second reviewer; conflicts were resolved by consensus, with input from additional reviewers if needed (BS, ASR, BF, FB, AO-A, or TH in English; TH or BH in German).

Data extraction and analysis

We extracted data corresponding to the proportion of individuals in the study cohorts who transitioned between diagnostic states for radiography, microbiology, and symptom status (figure 1) over a specified period of time. Diagnostic states were categorised as positive, negative, or mixed (ie, positive and negative or not specified). Recognising that descriptions of diagnostic states, and symptom status in particular, might not always be explicit by current standards, these could be recorded as unknown when microbiological status was clear. For studies that reported abnormal chest imaging that was suggestive of tuberculosis versus not suggestive of tuberculosis, we considered as abnormal and extracted data only for the tuberculosis-suggestive group. Additionally, we did not deem chest x-ray findings that were limited to calcified nodules to be abnormal for the purposes of this Review, on the basis of guidance indicating that patients with such findings require no intervention or follow-up.[18] We used the clinical classification method of the National Tuberculosis Association Diagnostic Standards and Classification of Tuberculosis to facilitate data extraction.[19] Baseline chest x-ray findings were further classified according to radiographic abnormalities reported by the original study authors as active, inactive, mixed (ie, active and inactive changes reported together or not specified), or negative (figure 2).

Some studies presented the proportion of individuals who progressed within a window of time rather than at a specific timepoint; in these cases, we present data points at the midpoint of the time window provided. To allow exploration of the data and any data heterogeneity, we attempted to collect data on variables of interest, as follows: age distribution, sex, frequency of follow-up visits, microbiological test used (culture *vs* smear), chest x-ray characteristics described by the study authors, tuberculin skin test data, local disease burden according to current WHO classification,[20] features of the study design (passive *vs* active *vs* mixed case finding and whether the data were generated from two cross-sectional assessments of participants [single follow-up] or through a cumulative count of events over time [cumulative count]), enrolment setting, and symptom status.

To allow comparison of progression across different cohort sizes and varying follow-up times, the proportions were standardised and the last data point in each cohort was annualised, enabling calculation of the expected number transitioning per year (appendix p 35). The variance of the annualised rate was then calculated using the escalc function from the Metafor package (version 3.8-1),[21] specifying the raw proportion measure. Meta-analysis was done using the rma function with the study outcome and variance as inputs. By default, each study was weighted proportional to the inverse of the variance calculated in the previous step. The forest plots were created using the forest function from the meta package (version 6.0-0). All summary estimates are presented with 95% CIs, calculated from the point data provided. Confidence interval proportions were limited to between 0 and 1 by the observation limit argument within the forest function. Subanalyses were also done using the rma function and added to the forest plot using the addpoly function from Metafor. Heterogeneity was assessed with the I^2 and τ^2 statistics. Prespecified sensitivity analyses were done to explore the effects of microbiological diagnostic modality (culture *vs* non-culture tests), with additional post-hoc sensitivity analyses guided by the data. Funnel plots were used to assess publication bias. These analyses with the above-mentioned packages were done with R (version 4.0.3).

Results

After removal of duplicates, titles and abstracts for a total of 10 623 studies were screened and full-text reports for 1794 studies were sought for retrieval (figure 3).[22–59] Full-text versions for 145 of the 1794 (8·1%) studies could not be sourced and 146 additional duplicates were identified. A further 1280 studies were deemed to meet exclusion criteria, leaving 223 for assessment of bias. A high risk of bias was found in 109 studies and data could not be extracted reliably from an additional 90, so these studies did not contribute to our results (appendix pp 25–29). In total, 22 English[22–29,31–39,41–59] and two German[30,40] studies, with a combined sample of 139 063 participants, contributed 34 cohorts for analysis. Eight of the 24 studies had maximum scores on the NOS general quality assessment; the quality of data on symptom status was

For the **Wellcome library** see https://wellcomecollection.org/pages/Wuw19yIAAK1Z3Smm

For the **British Library** see https://www.bl.uk

For more on the **German Central Committee against Tuberculosis** see https://www.dzk-tuberkulose.de

For the **German Tuberculosis Archive** see http://www.tb-archiv.de

For the **HathiTrust Digital Library** see https://www.HathiTrust.org

For the **US Internet Archive** see https://archive.org

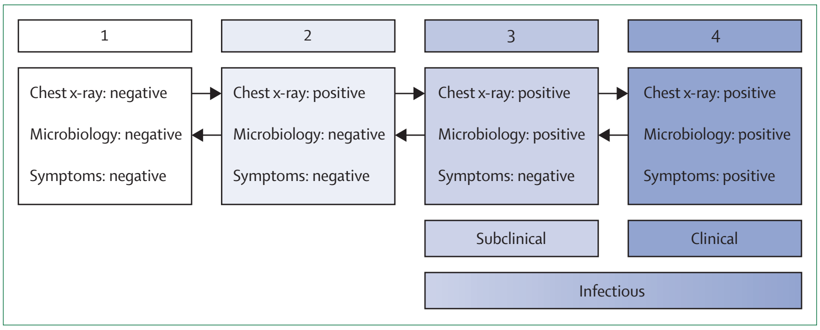
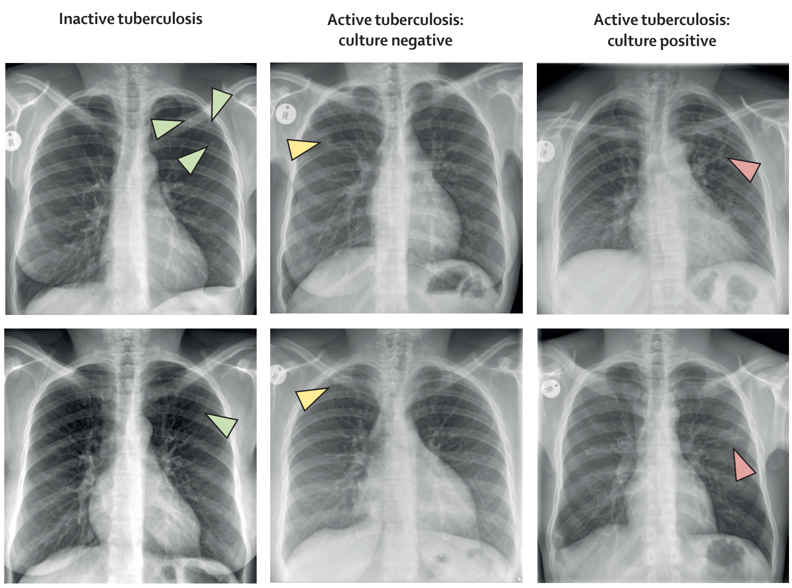


Figure 1: **Conceptual framework of transitions in the natural history of tuberculosis**
This conceptual framework, which is widely accepted but not published before in this form, is based on studies of the natural history of tuberculosis that include a subclinical group.[1,2,14–16] Individuals can fluctuate between the following states along the disease spectrum: (1) normal chest x-ray, negative microbiology (ideally tested with sputum culture or molecular testing), and asymptomatic; (2) chest x-ray abnormalities, negative microbiology, and asymptomatic; (3) chest x-ray abnormalities, positive microbiology, and asymptomatic; and (4) chest x-ray abnormalities, positive microbiology, and symptomatic. Individuals who have positive sputum microbiology, irrespective of symptom status, could be considered as potentially infectious. We recognise that individuals do not always fall into these groupings while progressing along this spectrum—eg, an individual might present with an abnormal chest x-ray and symptoms consistent with tuberculosis but have negative microbiology. In this Review, we have aimed to identify studies that include cohorts with all combinations of chest x-ray, microbiology, and symptom status.

A



B

Inactive tuberculosis
Discrete fibrotic scar or linear opacity Discrete linear or reticular opacity within the lung with or without volume loss
Discrete non-calcified nodule(s) One or more nodular opacities with distinct borders and no airspace consolidation
Other findings suggestive of previous tuberculosis eg, upper lobe bronchiectasis

Active tuberculosis
Infiltrate or consolidation Opacification of airspaces within the lung parenchyma
Cavitary lesion Lucency within the lung parenchyma that might be surrounded by airspace consolidation
Nodule with poorly defined margins Round opacity within the lung parenchyma
Pleural effusion Presence of fluid within the pleural space
Hilar or mediastinal lymphadenopathy Enlargement of lymph nodes within the hila, mediastinum, or both
Miliary nodules Nodules measuring 1–2 mm in size distributed throughout the lung parenchyma

Figure 2: **Chest x-ray findings in patients with active or inactive tuberculosis**
Tuberculosis can be associated with a range of abnormalities. However, parenchymal lesions will typically be within the upper zones of the lungs, predominately apically and sub-apically. (A) Chest x-rays from six individuals (≥18 years) showing representative abnormalities associated with inactive tuberculosis with no previous tuberculosis history (green arrows showing fibrotic scarring), active tuberculosis with negative culture (yellow arrows showing infiltrate), and active tuberculosis with positive culture (red arrows showing infiltrate). These chest x-rays were digital, from a recent active case-finding setting; all abnormalities were confirmed by CT scans. (B) Descriptions of lesions associated with active and inactive tuberculosis, based on the 2008 US Department of Health and Human Services technical instructions for medical examinations.[18] See appendix pp 21–23 for a description of imaging abnormalities used to distinguish active and inactive tuberculosis in the included studies.

generally poor, with 10 studies scoring no stars on the specific quality assessment (appendix p 9).

The settings for the 34 longitudinal cohorts were as follows: five workplace settings or university screening programmes, seven general community screening programmes, four household contact studies, nine clinical cohorts at clinics or sanatoria, and nine control groups in trials of therapeutic interventions (figure 4; appendix pp 10–20, 24). Ten cohorts were studied in Europe, eleven in Asia, eleven in North America, one in Africa, and one in South America. An estimate of the local burden of tuberculosis disease in the study setting and related time period was provided for 11 of the 34 cohorts (appendix p 24).[28–30,33,42–52,57] 9 of these 11 settings would be classified as endemic or high-burden tuberculosis settings[28,29,33,42–50,52,57] and the remaining two as medium-burden settings,[30,51] according to current WHO classification.[20] Cohorts were studied between 1923 and 2004, with 20 of 34 studied before 1960.

No cohorts meeting our inclusion and quality criteria reported the transition from normal chest x-ray to chest x-ray suggestive of tuberculosis in participants with confirmed recent conversion on tuberculin skin tests. We identified four cohorts that followed up participants with normal radiography and negative microbiological testing at baseline in which the timepoint of initial infection was unclear, with three reporting no evidence of symptoms and one with unrecorded symptom status (figure 4). 24 cohorts included follow-up of participants with evidence of radiographic abnormalities and negative microbiology at baseline; eight of these cohorts had no symptoms, three had symptoms, and 13 had mixed or unknown symptoms. Of these 24 cohorts, the radiographic abnormalities were reported by the original authors as active in nine cohorts, inactive or fibrotic in seven cohorts, and mixed or not specified in eight cohorts. Six cohorts followed participants with microbiologically detectable tuberculosis at baseline, of which four reported symptoms initially and symptom status was unknown in two; no cohorts in which patients were documented to be asymptomatic were identified. All six cohorts of participants with microbiologically detectable tuberculosis had abnormal chest x-rays at baseline. Study characteristics are summarised in figure 4. See the appendix for descriptions of chest x-ray findings (pp 21–23) and details of microbiological assessments and follow-up (pp 10–20) for each study.

Progression to positive microbiology in patients with negative microbiology at baseline

From the 24 cohorts with abnormal chest radiography but no evidence of *Mycobacterium tuberculosis* on respiratory sampling at baseline, representing 11 185 participants, development of microbiologically detectable incident disease occurred in between 1% (97 of 6990)[38] and 58% (88 of 152)[31] of individuals, with the studies reporting a median follow-up of 34·5 months (IQR 24–60; figure 5A). Considerable statistical heterogeneity was evident across cohorts (I^2=97·8%, τ^2=0·0028; p<0·0001). Funnel plots for the publications contributing to this primary analysis (appendix p 33) show that an asymmetrical distribution occurred for studies relating to inactive tuberculosis. We considered that the radiographic abnormalities categorised as active or inactive tuberculosis by the original authors (appendix pp 21–23) could represent distinct pathological states contributing to clinical variability across the studies. Therefore, we did not pool these studies in a meta-analysis, but instead conducted a stratified meta-analysis to describe the progression of these states separately. The annualised rate of transition from microbiologically negative to microbiologically positive disease was 10% (95% CI 6·2–13·3) for those in the nine cohorts described to have active changes on radiography and 1% (0·3–1·8) for those in the seven cohorts with inactive changes on radiography (figure 5A). Over a 3-year period, these annual rates would equate to 26% (17–35) of people with active tuberculosis changes versus 3% (1–5) of those with inactive tuberculosis changes progressing from microbiologically negative to positive disease. Statistical heterogeneity in the active and inactive tuberculosis subgroups was lower than in all cohorts taken together: I^2=77·4% (τ^2=0·0020; p<0·0001) and I^2=53·2% (τ^2<0·0001; p=0·051), respectively. The annual incidence in cohorts with mixed radiographic changes was 6% (1·5–11·1)—ie, between the values for inactive and active strata.

Of the 24 cohorts that contributed patients to this group, 18 used culture tests as part of the microbiological work-up and six did not specify the microbiological tests used (appendix pp 10–20). In a sensitivity analysis, restricting the cohorts to the 18 that used culture tests had little effect on the results (appendix p 30). Data on symptom status were provided for only 11 cohorts (figure 4). Of the nine cohorts described as having active tuberculosis changes on radiography, three[24,25,28,29] comprised symptomatic individuals (n=177). Progression in this subgroup was at an annualised rate of 12% (95% CI 2·7–20·8; appendix p 31). Only one cohort[52] was described as having active tuberculosis changes on radiography in an asymptomatic group, with symptom status unknown or mixed in the remaining five cohorts with active radiographic changes (figure 4).

In the four cohorts[25,42–50,52,57] that included follow-up of patients with no radiographic changes suggestive of tuberculosis (figure 4), transition to microbiologically positive disease occurred at an annualised rate of 0·1% (95% CI 0·1–0·2; not shown). In single follow-up and cumulative count studies, patients with active tuberculosis changes showed a similar rate of annual progression (appendix p 32).

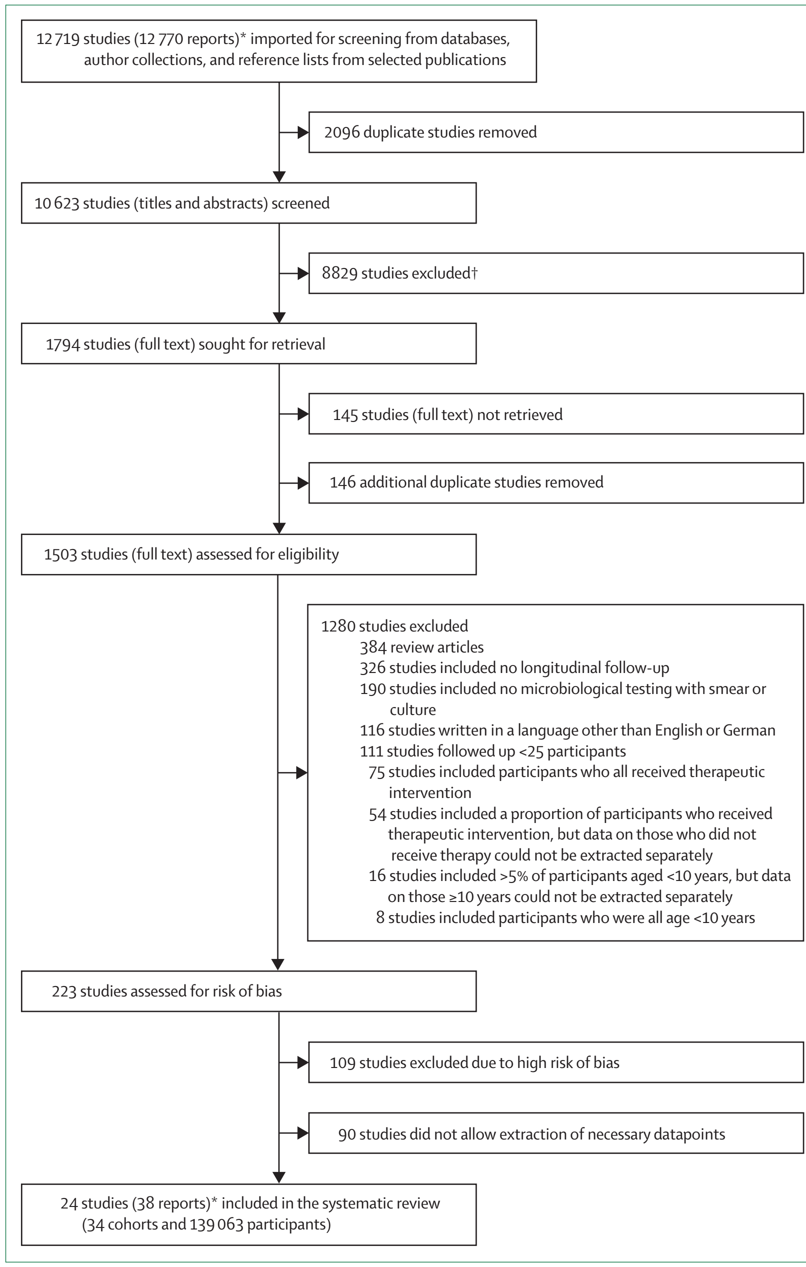


***Figure 3:* Study selection**
*The number of reports is higher than the number of studies; during the study selection process, publications related to the same study were merged, either manually or by the Covidence systematic review software. The final output was checked manually and 38 reports were found to be associated with the 24 studies. †Studies were excluded if they were not deemed to be relevant to the aims of this Review.

Regression to negative microbiology in patients with positive microbiology at baseline

Six cohorts included follow-up of a total of 1024 participants with evidence of *M tuberculosis* in respiratory samples at baseline and assessed the proportion transitioning to a microbiologically undetectable state without treatment or intervention (figure 4). The median follow-up for the cohorts was 34·5 months (IQR 13–54; figure 5B). Most of these cohorts included participants with minimal disease on chest x-ray, owing to the entry criteria of the original study or to the eligibility criteria of this systematic review. No studies adequately described symptom status of the participants and all were done before the discovery of HIV. Of the six cohorts,[25,32,40–50,55] three were retrospective cohorts from tuberculosis hospitals or sanatoria, and three were prospective cohorts from the general community or household surveys or from the control arm of a clinical trial. In four of the six cohorts, culture tests were used to

	Study type	Years of study or publication*	Age of participants	Cohort size (n)	Baseline radiographic classification†	Type of follow-up‡	Starting point§	Endpoint§
Alling et al (1955; USA)[22]	Retrospective cohort (clinic, hospital, or sanatorium)	1938–48	Mean 52 years	58	Inactive	Cumulative	Chest x-ray: positive Microbiology: negative Symptoms: unknown (arrested¶)	Chest x-ray: positive Microbiology: positive Symptoms: unknown
Anastasatu et al (1985; Romania)[23]	Control (no treatment or placebo) arm of prospective clinical trial	Published 1985	Not reported	143	Inactive	Cumulative	Chest x-ray: positive Microbiology: negative Symptoms: negative	Chest x-ray: positive Microbiology: positive Symptoms: unknown
Aneja et al (1979; India)[24]	Control (no treatment or placebo) arm of prospective clinical trial	1975–77	≥12 years	110	Active	Single	Chest x-ray: positive Microbiology: negative Symptoms: positive	Chest x-ray: positive Microbiology: positive Symptoms: unknown
Beeuwkes et al (1942; USA)[25]	Prospective cohort (household contact study); four subgroups based on chest x-ray lesion type and sputum microbiology	1933–38	Not reported	784	Negative	Single	Chest x-ray: negative Microbiology: negative Symptoms: negative	Chest x-ray: unknown Microbiology: positive Symptoms: positive
				79	Mixed	Single	Chest x-ray: positive Microbiology: negative Symptoms: negative	Chest x-ray: unknown Microbiology: positive Symptoms: positive
				43	Active	Single	Chest x-ray: positive Microbiology: negative Symptoms: positive	Chest x-ray: unknown Microbiology: positive Symptoms: positive
				28	Active	Single	Chest x-ray: positive Microbiology: positive Symptoms: positive	Chest x-ray: unknown Microbiology: negative Symptoms: unknown
Bobrowitz et al (1947, 1949; USA)[26,27]	Prospective cohort (clinic, hospital, or sanatorium)	1938–45	Not reported	191	Mixed	Single	Chest x-ray: positive Microbiology: negative Symptoms: unknown	Chest x-ray: positive Microbiology: positive Symptoms: unknown
Borgen et al (1950, 1951; Norway)[28,29]	Prospective cohort (occupational or student screening); two subgroups based on symptoms	1947–49	≥15 years	24	Active	Single	Chest x-ray: positive Microbiology: negative Symptoms: positive	Chest x-ray: positive Microbiology: positive Symptoms: positive
				120	Inactive	Single	Chest x-ray: positive Microbiology: negative Symptoms: negative	Chest x-ray: positive Microbiology: positive Symptoms: positive
Breu (1954; Germany)[30]	Prospective cohort (general community survey)	1949–52	≥15 years	904	Mixed	Single	Chest x-ray: positive Microbiology: negative Symptoms: unknown	Chest x-ray: positive Microbiology: positive Symptoms: unknown
Cowie et al (1985; South Africa)[31]	Prospective cohort (occupational or student screening	1979–84	Not reported	152	Active	Cumulative	Chest x-ray: positive Microbiology: negative Symptoms: unknown	Chest x-ray: positive Microbiology: positive Symptoms: unknown
Downes (1938; USA)[32]	Retrospective cohort (clinic, hospital, or sanatorium)	1923–35	Range 15–69 years	342	Active	Cumulative	Chest x-ray: positive Microbiology: positive Symptoms: positive	Chest x-ray: positive Microbiology: negative Symptoms: negative
Frimodt-Moller et al (1965; India)[33]	Control (no treatment or placebo) arm of prospective clinical trial	1960–61	≥15 years	86	Active	Cumulative	Chest x-ray: positive Microbiology: negative Symptoms: unknown	Chest x-ray: positive Microbiology: positive Symptoms: unknown
Hong Kong Chest Service (1979, 1981, 1984; Hong Kong)[34–37]	Control (no treatment or placebo) arm of prospective clinical trial	Published 1979, 1981, 1984	Range 15–75	176	Active	Cumulative	Chest x-ray: positive Microbiology: negative Symptoms: mixed	Chest x-ray: positive Microbiology: positive Symptoms: unknown
IUAT Committee on Prophylaxis (1982; Europe)[38]	Control (no treatment or placebo) arm of prospective clinical trial	Published 1982	Mean 50 years	6990	Inactive	Cumulative	Chest x-ray: positive Microbiology: negative Symptoms: unknown	Chest x-ray: positive Microbiology: positive Symptoms: unknown

Patients with radiologically and microbiologically negative findings
Patients with radiological abnormalities and microbiologically negative findings
Patients with confirmed microbiologically positive disease

(Figure 4 continues on next page)

assess the microbiological status of the participants, whereas in two cohorts, both retrospective, either microscopy was used or the nature of the microbiological investigations was not specified (appendix pp 10–20). The meta-analysis showed that this transition occurred at an annualised rate of 18% (95% CI 3·0–33·7; figure 5B), but there was considerable heterogeneity across studies (I^2=98·1%, τ^2=0·033; p<0·0001). We then restricted the

	Study type	Years of study or publication*	Age of participants	Cohort size (n)	Baseline radiographic classification†	Type of follow-up‡	Starting point§	Endpoint§
Lincoln et al (1954; USA)[39]	Retrospective cohort (clinic, hospital, or sanatorium)	1937–47	≥14 years; mean 24 years	314	Inactive	Cumulative	Chest x-ray: positive Microbiology: negative Symptoms: unknown (arrested¶)	Chest x-ray: positive Microbiology: positive Symptoms: unknown
Manser (1953; Switzerland)[40]	Retrospective cohort (clinic, hospital, or sanatorium)	1941–51	Range 60–83 years	40	Active	Single	Chest x-ray: positive Microbiology: positive Symptoms: unknown	Chest x-ray: positive Microbiology: negative Symptoms: unknown
Marshall et al (1948; UK)[41]	Control (no treatment or placebo) arm of prospective clinical trial	1947–48	Range 15–30 years	52	Active	Cumulative	Chest x-ray: positive Microbiology: positive Symptoms: positive	Chest x-ray: positive Microbiology: negative Symptoms: unknown
National Tuberculosis Institute (1974, 1976, 1978, 1982; India)[42–50]	Prospective cohort (general community survey); three subgroups based on chest x-ray lesion type and sputum microbiology	1961–68	≥5 years	31490	Negative	Single	Chest x-ray: negative Microbiology: negative Symptoms: unknown	Chest x-ray: positive Microbiology: positive Symptoms: unknown
				271	Active	Single	Chest x-ray: positive Microbiology: negative Symptoms: unknown	Chest x-ray: positive Microbiology: positive Symptoms: unknown
				178	Active	Single	Chest x-ray: positive Microbiology: positive Symptoms: unknown	Chest x-ray: positive Microbiology: negative Symptoms: unknown
Nørregaard et al (1990; Denmark)[51]	Control (no treatment or placebo) arm of prospective clinical trial	1978–85	≥20 years	28	Active	Cumulative	Chest x-ray: positive Microbiology: negative Symptoms: mixed	Chest x-ray: positive Microbiology: positive Symptoms: negative
								Chest x-ray: positive Microbiology: positive Symptoms: positive
Okada et al (2012; Cambodia)[52]	Retrospective cohort (general community survey); two subgroups based on chest x-ray lesion type and sputum microbiology	2002–04	≥10 years; median 30·6 years	309	Active	Single	Chest x-ray: positive Microbiology: negative Symptoms: negative	Chest x-ray: positive Microbiology: positive Symptoms: unknown
								Chest x-ray: positive Microbiology: positive Symptoms: positive
								Chest x-ray: negative Microbiology: negative Symptoms: unknown
				21580	Negative	Single	Chest x-ray: negative Microbiology: negative Symptoms: negative	Chest x-ray: positive Microbiology: positive Symptoms: unknown
Orrego Puelma and Grebe (1945; Chile)[53]	Retrospective cohort (clinic, hospital, or sanatorium)	Published 1945	≥15 years	67	Mixed	Single	Chest x-ray: positive Microbiology: negative Symptoms: unknown	Chest x-ray: positive Microbiology: positive Symptoms: unknown
Pamra and Mathur (1971; India)[54]	Control (no treatment or placebo) arm of prospective clinical trial	1958–68	Range 15–45 years	178	Mixed	Cumulative	Chest x-ray: positive Microbiology: negative Symptoms: negative	Chest x-ray: positive Microbiology: positive Symptoms: positive
								Chest x-ray: positive Microbiology: positive Symptoms: negative

(Figure 4 continues on next page)

	Study type	Years of study or publication*	Age of participants	Cohort size (n)	Baseline radiographic classification†	Type of follow-up‡	Starting point§	Endpoint§
Puffer et al (1945; USA)[55]	Retrospective cohort (clinic, hospital, or sanatorium); three subgroups based on chest x-ray lesion type and sputum microbiology	1931–43	Not reported	261	Mixed	Single	Chest x-ray: positive Microbiology: negative Symptoms: negative	Chest x-ray: positive Microbiology: positive Symptoms: positive
				267	Inactive	Single	Chest x-ray: positive Microbiology: negative Symptoms: negative (arrested¶)	Chest x-ray: positive Microbiology: positive Symptoms: positive
				384	Active	Single	Chest x-ray: positive Microbiology: positive Symptoms: positive	Chest x-ray: positive Microbiology: negative Symptoms: negative
Sikand et al (1959; India)[56]	Prospective cohort (occupational or student screening)	1952–58	≥15 years	167	Inactive	Cumulative	Chest x-ray: positive Microbiology: negative Symptoms: unknown	Chest x-ray: positive Microbiology: positive Symptoms: unknown
				152	Mixed	Cumulative	Chest x-ray: positive Microbiology: negative Symptoms: unknown	Chest x-ray: positive Microbiology: positive Symptoms: unknown
Stýblo et al (1967; Czechoslovakia)[57]	Prospective cohort (general community survey)	1961–65	≥15 years	73 000	Negative	Cumulative	Chest x-ray: negative Microbiology: negative Symptoms: negative	Chest x-ray: unknown Microbiology: positive Symptoms: positive
								Chest x-ray: unknown Microbiology: positive Symptoms: unknown
Tuberculosis Society of Scotland (1958),[58] Scottish Thoracic Society (1963)[59] (Scotland)	Control (no treatment or placebo) arm of prospective clinical trial	1954–59	≥15 years	95	Mixed	Single	Chest x-ray: positive Microbiology: negative Symptoms: negative	Chest x-ray: positive Microbiology: positive Symptoms: unknown

Figure 4: **Study characteristics**

IUAT=International Union Against Tuberculosis. *Publication year(s) given if study dates not reported. †Classified according to abnormalities on chest x-ray reported by the original study authors as active, inactive, mixed, or negative. ‡Single follow-up refers to studies with two cross-sectional assessments of the participant groups; cumulative follow-up refers to studies that cumulatively captured events over time. §Starting points and endpoints have three characteristics or states, including radiology (ie, chest x-ray negative, positive, or unknown), microbiology (ie, negative or positive), and symptom status (ie, negative, positive, unknown, or mixed). ¶Group known to have been microbiologically positive on a prior occasion, then documented arrested disease, followed by a relapse.

meta-analysis to prospective studies (hence removing the three hospital or sanitoria cohorts, including the two that had not used culture tests);[25,41–50] this resulted in an annualised rate of 12% (6·8–18·0) with reduced statistical heterogeneity I^2=35·1% (τ^2=0·0008; p=0·25). Over 3 years, this proportion would equate to 33% (19–45) of patients initially with culture-positive tuberculosis converting to culture-negative disease.

Discussion

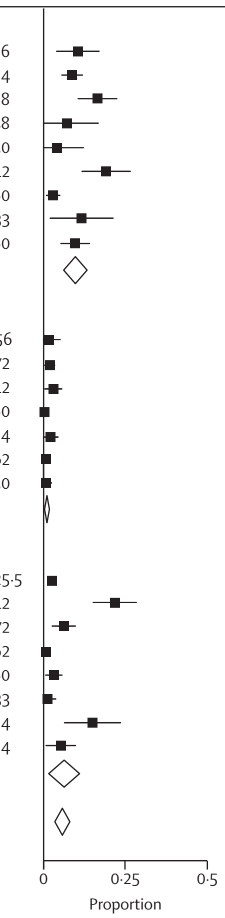
Quantifying progression and regression in pulmonary disease

To our knowledge, this Review is the first to systematically summarise key aspects of the kinetics of the natural history of untreated tuberculosis in adults, outside of mortality, making full use of studies published in English and German. Through meta-analyses, we provide estimates of the risk of progression to microbiologically positive disease in patients who initially had negative microbiology, with an annualised rate of approximately 10% in those with active radiographic tuberculosis changes and 1% in those with inactive or fibrotic changes. For comparison, progression was approximately 0·1% in patients with normal chest x-ray at baseline, although this rate would be affected by factors such as the local burden of disease. Additionally, we provide an estimate for the reversion from culture-positive disease to culture-negative disease without treatment (also referred to as self-cure) at 12% per year.

Our results show that individuals with chest x-ray changes suggestive of active tuberculosis who are initially negative on microbiological tests are at considerable risk of disease progression, with 26% predicted to progress over 3 years. Our analysis is the first to provide an estimate for this transition, which could be used by modellers aiming to understand the implications of intervention use in this population. We have also shown that about a third of patients with culture-positive disease could revert to culture-negative disease without treatment over a 3-year period. Although this might not inform clinical management, our results could be used to refine parameters in models to estimate disease incidence from prevalence survey data that need to account for the probability of so-called self-cure. The annual rate of about 12% provides an empirical foundation for the slightly

A

	Patients who progressed (n)	Cohort size (n)	Follow-up (months)	Annualised rate (95% CI)
Active				
Frimodt-Moller et al (1965)[33]	25	86	36	0·10 (0·04–0·17)
Okada et al (2012)[52]	51	309	24	0·09 (0·06–0·12)
Cowie et al (1985)[31]	88	152	58	0·16 (0·11–0·22)
Nørregaard et al (1990)[51]	8	28	48	0·07 (0·00–0·17)
Borgen et al (1950, 1951)[28,29]	2	24	30	0·04 (0·00–0·12)
Aneja et al (1979)[24]	21	110	12	0·19 (0·12–0·26)
National Tuberculosis Institute (1974, 1976, 1978, 1982)[42–50]	36	271	60	0·03 (0·01–0·05)
Beeuwkes et al (1942)[25]	13	43	33	0·12 (0·02–0·21)
Hong Kong Chest Service (1979, 1981, 1984)[34–37]	71	176	60	0·10 (0·05–0·14)
Random-effects model				**0·10 (0·06–0·13)**
Heterogeneity: Q=40·8, df=8 (p<0·0001); I^2=77·4%, τ^2=0·0020				
Inactive				
Alling et al (1955)[22]	10	58	156	0·02 (0·00–0·05)
Lincoln et al (1954)[39]	36	314	72	0·02 (0·00–0·03)
Sikand et al (1959)[56]	5	167	12	0·03 (0·00–0·06)
IUAT Committee on Prophylaxis (1982)[38]	97	6990	60	0·00 (0·00–0·00)
Anastasatu et al (1985)[23]	6	143	24	0·02 (0·00–0·04)
Puffer et al (1945)[55]	10	267	62	0·01 (0·00–0·02)
Borgen et al (1950, 1951)[28,29]	2	120	30	0·01 (0·00–0·02)
Random-effects model				**0·01 (0·00–0·02)**
Heterogeneity: Q=12·6, df=6 (p=0·051); I^2=53·2%, τ^2<0·0001				
Mixed				
Breu (1954)[30]	48	904	25·5	0·03 (0·02–0·04)
Sikand et al (1959)[56]	33	152	12	0·22 (0·15–0·28)
Pamra and Mathur (1971)[54]	57	178	72	0·06 (0·03–0·10)
Puffer et al (1945)[55]	9	261	62	0·01 (0·00–0·02)
Bobrowitz et al (1947, 1949)[26,27]	26	191	60	0·03 (0·01–0·06)
Beeuwkes et al (1942)[25]	3	79	33	0·01 (0·00–0·04)
Orrego Puelma and Grebe (1945)[53]	18	67	24	0·15 (0·06–0·23)
Tuberculosis Society of Scotland (1958),[58] Scottish Thoracic Society (1963)[59]	9	95	24	0·05 (0·01–0·10)
Random-effects model				**0·06 (0·02–0·11)**
Heterogeneity: Q=57·8, df=7 (p<0·0001); I^2=97·3%, τ^2=0·0041				
Random-effects model for all studies				**0·06 (0·03–0·08)**
Heterogeneity: Q=223·9, df=23 (p<0·0001); I^2=97·8%, τ^2=0·0028				
Test for subgroup differences: Q_M=12·6, df=2 (p=0·0019)				



B

	Patients who regressed (n)	Cohort size (n)	Follow-up (months)	Annualised rate (95% CI)
Prospective				
Marshall et al (1948)[41]	2	52	6	0·08 (0·00–0·15)
National Tuberculosis Institute (1974, 1976, 1978, 1982)[42–50]	70	178	36	0·15 (0·10–0·20)
Beeuwkes et al (1942)[25]	10	28	33	0·14 (0·01–0·27)
Random-effects model				**0·12 (0·07–0·18)**
Heterogeneity: Q=2·7, df=2 (p=0·25); I^2=35·1%, τ^2=0·0008				
Retrospective				
Manser (1953)[40]	15	40	6	0·60 (0·45–0·75)
Downes (1938)[32]	171	342	60	0·13 (0·09–0·16)
Puffer et al (1945)[55]	92	384	62	0·05 (0·03–0·07)
Random-effects model				**0·25 (–0·08 to 0·58)**
Heterogeneity: Q=58·3, df=2 (p<0·0001); I^2=99·5%, τ^2=0·082				
Random-effects model for all studies				**0·18 (0·03–0·33)**
Heterogeneity: Q=65·1, df=5 (p<0·0001); I^2=98·1%, τ^2=0·033				
Test for subgroup differences: Q_M=0·5, df=1 (p=0·46)				

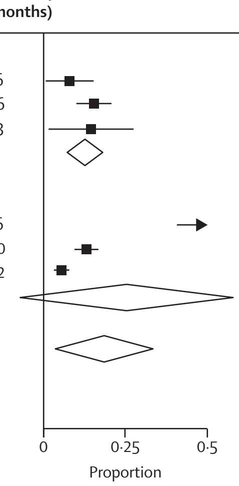


Figure 5: **Rates of transition from bacteriologically negative to bacteriologically positive disease and bacteriologically positive to bacteriologically negative disease**

(A) Forest plot of the random-effects meta-analysis of cohorts with abnormal chest x-rays and negative microbiology that transitioned to positive microbiology during follow-up, stratified by initial radiographic classification. Numbers who transitioned in each cohort with annualised rates of progression and 95% CIs are shown for radiographic subgroups. Subgroups are classified according to baseline radiographic abnormalities, reported by the original study authors as active, inactive, mixed, or negative. (B) Forest plot of the random-effects meta-analysis of cohorts with positive microbiology that transitioned to negative microbiology during follow-up, stratified by study design. Numbers who transitioned in each cohort with annualised rates of regression and 95% CIs are shown for prospective and retrospective study designs. Values for annualised rate (95% CI) are given to two decimal places.

higher rates of 15% and 20% used by Dye and colleagues to estimate parameters for self-cure, which were informed by a (non-systematic) literature review.[60,61] However, importantly, the patients included in our systematic review might not be representative of all culture-positive patients, as our focus was on more minimal disease.

We used a widely accepted conceptual framework to guide data collection, which required determination of the microbiological, radiological, and symptom status of participants over follow-up. We found that no single study systematically recorded these three features over the course of disease from exposure to final outcome. Additionally, recording of symptoms in these studies was not explicit, particularly during follow-up, so we had insufficient empirical data to determine directly the trajectory of subclinical (asymptomatic, microbiologically positive) tuberculosis. Subclinical tuberculosis is a state commonly identified through chest x-ray-based active case finding, but conducting contemporary natural history studies to determine the rates of progression and regression would present ethical challenges now that treatments for tuberculosis are available. However, the substantial additional data reported in this Review should allow inferences about the kinetics of subclinical tuberculosis, which Richards and colleagues have explored in a modelling study using a Bayesian framework to consider all data simultaneously and also incorporating additional available evidence on subsequent mortality.[62] The modelling work suggests that for individuals with prevalent subclinical disease, classic clinical disease is neither an inevitable nor an irreversible outcome. At 5 years, 40% (95% uncertainty interval 31·3–48·0) had recovered but 18% (13·3–24·0) had died from tuberculosis, with 14% (9·9–19·2) still infectious.[62] Furthermore, 50% (40·0–59·1) of the subclinical cohort never developed symptoms over the model span.[62] Overall, these results suggest that a reliance on symptom-based screening means that a large proportion of people with infectious disease might never be detected.

There are several key limitations to consider when interpreting the findings of this systematic review. First, HIV has a considerable role in the epidemiology of tuberculosis in certain settings, and 22 of the 24 included studies were done before the discovery of the virus. It is likely that people living with HIV progress along the disease spectrum with different kinetics, which could also be influenced by immune status.[63–65] Second, the nature of this research question and the historical focus resulted in studies being included from a period spanning almost 80 years; over this time, microbiological and radiographic methods evolved (appendix p 34). However, from a microbiological perspective, most included studies used culture tests, and sensitivity analyses in which studies that did not use culture tests were excluded showed similar results to those with the overall cohort. From a radiological perspective, findings from studies that used mass miniature radiography or fluoroscopy for screening[28,29] were typically confirmed with conventional chest radiography, which informed data extraction. Most studies were done more than 50 years ago, when socioeconomic factors, health-care access, comorbidity distribution, and prevalence of tuberculosis were likely to have differed from those of today, and could have affected the rate of progression and regression of disease. However, these study environments might to some extent remain representative of many contemporary settings with a high tuberculosis burden.

Furthermore, although we allowed for data capture along multiple possible trajectories of tuberculosis disease, it is possible that we did not capture all the various possible trajectories that exist along this pathway. Although data were not available for certain variations (eg, starting with microbiologically detectable tuberculosis but a radiographically normal state), availability of data would have been affected by the designs of the included studies, but could also have been affected by the diagnostic methods used—ie, chest x-ray rather than more modern and sensitive methods. Our findings could also be affected by publication bias, as shown by the asymmetrical funnel plot (appendix p 33); this appears to be relevant mainly for studies of inactive tuberculosis, suggesting that small studies with no transitions might not have been published. Additionally, some studies could have had a survival bias because they required participants to meet certain entry criteria that were stable over time. Our results on progression to microbiological positivity in individuals with baseline chest x-ray abnormalities are drawn from studies with a median follow-up of 34·5 months (approximately 3 years; IQR 24–60) and the annualised rates that we report are not expected to apply outside of this time period. Our transitions reflect the course of disease in patients who were followed up and successfully provided sputum for microbiological analysis (not accounting for death and loss to follow-up), and hence it is possible that the true rates could be higher. Importantly, progression to microbiologically positive disease from a microbiologically undetectable state could represent disease progression or new, incident infection and disease—a factor that is likely to be affected by the local burden of disease.

There are further considerable methodological challenges in conducting a systematic review involving historical research. It is notable that 1503 of 1648 (91·2%) studies (excluding duplicates) were retrieved for full-text review; however, for 90 studies that met the eligibility and bias criteria, manuscript style did not allow data extraction and the original authors could not be contacted for assistance. Although our work focused on the period from 1903 to 1960, through searches of extensive investigator collections and identification of further references from selected articles, we are confident that we were able to identify key studies that included untreated cohorts published after 1960, as evidenced by the fact that reports of nearly half of the final 24 studies were published after this date.

Contemporary approaches to understanding disease natural history

Our study highlights the fact that infiltrative pathology can be evident on chest x-ray before sputum positivity, that progression to sputum positivity can take months or years, and that risk can be stratified by features of activity on chest x-ray. We also show that in those with positive sputum, reversion to a sputum-negative state can occur. This work reiterates to a modern-day audience the chronic and dynamic natural history of tuberculosis that would have been more apparent to researchers and physicians historically. As discussed, the approaches used in historical studies have limitations compared with modern-day methods. However, in contemporary studies we can examine disease natural history in humans only until the point at which treatment is clinically indicated. Digital chest x-ray technologies are now commonplace and computer-aided detection software enables more consistent and highly sensitive reading of chest x-rays.[66] Chest x-rays are limited in their anatomical resolution, with visibility of underlying lesions affected by their size, location, and density (figure 2). In studies using CT or PET-CT, earlier stages of disease can be visualised, with centrilobular nodules representing caseous and cellular material within the respiratory bronchioles, which grow and coalesce to form denser consolidation that might be visible on chest x-ray.[63,67,68]

Similarly, sputum investigation has limitations because it requires organisms from the site of disease to enter respiratory secretions and to be effectively expectorated as sputum. Also, sputum assessments are relatively infrequent in almost all studies and hence cannot easily be used to capture variation in sputum positivity over short time periods. Tuberculosis transmission occurs through aerosols and it is becoming increasingly apparent that capture of aerosols (eg, through face-mask sampling) might be more sensitive than sputum microbiology and might also better reflect infectiousness.[69] Additionally, as discussed, historical studies did not capture information about symptoms effectively, especially over follow-up. Incorporating these newer detection methods into modern epidemiological studies might help to address key outstanding research questions (panel).

Furthermore, the host–pathogen interplay that governs the dynamic nature of the tuberculosis disease course and the factors that could lead to a favourable or unfavourable outcome are poorly understood. This interplay cannot easily be studied in humans, but could be addressed with animal models. Traditionally, animal models of tuberculosis have aimed to replicate the formation of the granuloma but not specific stages of early disease evolution. More accurate benchmarking of animal models against the early stages of tuberculosis disease will facilitate progress towards a better understanding of factors that govern disease outcome.[70]

Future directions for disease management

Our finding of a 10% annualised rate of disease progression in patients with negative sputum microbiology and chest x-ray changes suggestive of active tuberculosis indicates that although this group is at high risk of progression, this is not inevitable. These individuals are still frequently encountered in two clinical settings. First, such individuals could be found in the context of active case finding in which a target population not seeking health care is screened with chest x-ray; this population is increasingly being recognised following recent WHO guidance on systematic screening for tuberculosis, which recommends the use of chest x-ray.[66] Second, these individuals could present as patients who are symptomatic and seeking health care, with negative sputum investigation but who are found to have chest x-ray abnormalities. The optimal approach to management of this group is currently unclear, particularly in resource-limited programmatic settings where investigations such as CT scanning and bronchoscopy are not routinely available. Treatment algorithms vary widely, but ultimately rely on clinician judgement to factor in symptoms, epidemiological risk, and the likelihood of multidrug resistance, with the tension between the need to provide empirical treatment versus monitoring potentially leading to over-treatment or under-treatment. Recent clinical trials in this patient group are limited and the current one-size-fits-all approach typically means that the standard 6-month, four-drug therapy developed for the treatment of

Panel: Key questions for future research

- In patients with normal chest x-ray and disease-positive sputum microbiology, can pathological changes be identified in the lung using higher-resolution imaging?
- Using modern digital chest x-ray technologies with computer-aided detection, are the rates of disease progression similar to those found in the historical literature?
- Do the rates of progression and regression across the disease spectrum differ according to symptom status?
- In patients with chest x-ray changes suggestive of tuberculosis but with negative sputum microbiology, is the rate of progression to positive sputum microbiology constant over time?
- What proportion of patients with chest x-ray changes suggestive of tuberculosis but with negative sputum microbiology have microbiologically positive aerosol (eg, with face-mask sampling)?
- Is there evidence for transmission of tuberculosis in patients with chest x-ray changes but negative sputum microbiology?
- To what extent does sputum positivity vary in individuals with tuberculosis over short time periods?
- Can current or novel diagnostic tests (eg, C-reactive protein and host transcriptional response tests) help to identify patients with chest x-ray changes at risk of microbiological progression?
- What is the optimal therapeutic approach to prevent progression to microbiologically positive disease in patients with chest x-ray changes suggestive of tuberculosis?

smear-positive disease is offered to these patients with minimal disease.[71] New approaches are needed to support the management of this group. Novel diagnostics that could provide either microbiological confirmation (eg, face-mask sampling) or better risk stratification (eg, C-reactive protein or host transcriptional response tests) in a clinical setting require further evaluation.[69,72] Additionally, clinical trials are needed to evaluate preventive treatment strategies that are better tolerated and to determine the number needed to treat to improve patient choice and facilitate clinical decision making.[71]

Conclusion

In this Review, we have shown that the natural history of tuberculosis is a dynamic, heterogeneous process that is not adequately represented by a single active disease state, and we have quantified three key transitions: progression from bacteriologically negative to bacteriologically positive disease in individuals with chest x-ray changes suggestive of active disease and in those with chest x-ray changes suggestive of inactive disease, and regression from bacteriologically positive to bacteriologically negative disease. Moreover, we have provided a much-needed foundation of empirical data for our ongoing rediscovery of the complexity of tuberculosis natural history, enabling the development and testing of new hypotheses and a drive towards new clinical guidelines and policies for the prevention and management of tuberculosis disease.

Contributors

HE, RMGJH, BS, ASR, FC, and KK conceptualised the study protocol, with additional scientific input from AO. BS, ASR, TH, BF, FB, AO-A, and BH did the literature search and data collection. ASR and BS accessed and verified the data, and did the statistical analysis with input and oversight from HE, RMGJH, and ER. BS wrote the first draft of the manuscript with input from ASR, RMGJH, and HE. All authors subsequently reviewed and edited the manuscript. All authors had full access to the study data and had final responsibility for the decision to submit for publication.

Declaration of interests

We declare no competing interests.

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